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Palaeosecular variation in Central Mexico over the last 30 000 years: the record from lavas

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SUMMARY

13 lava flows of known age (ages from ^{14}C dating), which have been erupted in the last 30 000 years, have been studied to determine the palaeosecular variation of the geomagnetic field in Central Mexico. Samples were taken from two different monogenetic volcanic fields: the Michoacan–Guanajuato volcanic field (six sites) and the Chichinautzin Formation (seven sites), both part of the Transmexican Volcanic Belt. The lavas were studied in detail using rock magnetic methods (magnetic susceptibility at room temperature, low-temperature susceptibility behaviour, hysteresis loops, Curie temperatures), combined with reflected light microscopy, in order to deduce their magnetic mineralogy and the domain states of the magnetic minerals. The magnetic carriers are titanomagnetites, which show differing degrees of high-temperature deuteric oxidation, and seem to be predominantly pseudo-single domain (PSD), though in many cases are probably a mixture of domain states. Mean palaeomagnetic directions and palaeointensity values using Shaw and Thellier techniques were obtained using several specimens from each flow. Our data seem to indicate a sharp easterly swing in declination about 5000 years ago, which is also observed in lake sediments from Central Mexico. The calculated values of the virtual dipole moment (VDM) range from 3.1 to 14.9×10^{22} A m^2. Our data indicate that the virtual dipole moment seems to have increased gradually in magnitude over the last 30 kyr, with a peak at about 9000 years BP. These are features that have been observed in other parts of the globe and are probably caused by variations in the dipole part of the geomagnetic field.

Key words: magnetic mineralogy, Mexico, palaeointensity, palaeomagnetism, Quaternary, secular variation.

INTRODUCTION

For the late Quaternary, studies of palaeosecular variation from lake sediments, lavas and archaeological material are fairly plentiful from Europe, the USA and Japan, but very limited for the rest of the world, especially the regions with latitudes less than 30° in the Northern Hemisphere (e.g. Mexico) and the Southern Hemisphere. A better geographical coverage is required to refine models of geomagnetic field behaviour. Knowledge of the spatial and temporal variations of the palaeosecular variation may allow us to estimate the topography and characteristics of the core–mantle interface and lateral uniformity in the lower mantle (e.g. Doell & Cox 1972; McFadden & Merrill 1995).

During historical times, on the basis of direct observations of the geomagnetic field, it has been observed that the direction as well as the magnitude of the non-dipolar field, which produces the majority of the observed secular variation, has a very irregular distribution over the Earth's surface. This non-dipolar component is at the present time larger in the Southern Hemisphere than the Northern Hemisphere, and is almost non-existent in the Central Pacific region (Doell & Cox 1972). Analyses of late Quaternary palaeomagnetic results from Hawaii have tended to show a lower than predicted angular dispersion of virtual geomagnetic poles (VGPs), both from lavas (Doell & Cox 1972; McWilliams, Holcomb & Champion 1982) and from sedimentary sequences (Peng & King 1992). Detailed comparison of virtual dipole moments (Smith 1967) calculated using palaeointensities from ^{14}C-dated Hawaiian lavas (Coe, Grommé & Mankinen 1978; Tanaka & Kono 1991; Mankinen & Champion 1993a) with data from the rest of the

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world has indicated that the picture is more complex and that strong dipole fields were present in Hawaii between 5000 and 200 years BP, but absent between about 12 000 and 5000 years BP (Mankinen & Champion 1993b).

Most previous studies of palaeosecular variation in Mexico use lists of Quaternary and late Tertiary palaeomagnetic data to determine the mean magnitude of palaeosecular variation in this region. Herrero-Bervera *et al.* (1986) believe that Mexico is not included in the 'dipolar window' of low values of palaeosecular variation in which Hawaii lies. However, a later analysis by Böhnel, Urrutia Fucugauchi & Herrero-Bervera (1990) found that the non-dipolar component is lower that that expected for this latitude; they propose the existence of a 'belt' of low or almost non-existent values of palaeosecular variation that includes Hawaii and Mexico. The apparent discrepancy between the two studies seems to have been produced by the different models used to estimate the magnitude of secular variation.

The only study that produced a palaeosecular variation record from Mexico is that of Latham, Schwartz & Ford (1986), who described the results from a 1200-year-old stalagmite from NE Mexico, dated by the U–Th method. Obviously, with such a short record it is difficult to say a great deal about the secular variation in Mexico.

Our principal objective was to obtain preliminary reference curves for the palaeosecular variation (declination, inclination and intensity) in Central Mexico for the last 30 000 years using the record preserved by lavas for which ^{14}C ages exist. At the same time another group from UNAM were trying to obtain a record of palaeosecular variation from late Quaternary lacustrine sediments from Lake Chalco, close to Mexico City (Ortega-Guerrero 1992; Urrutia Fucugauchi *et al.* 1995).

We present the results of a detailed study of geomagnetic secular variation recorded by lavas from two separate volcanic fields in the Transmexican Volcanic Belt, located in Central Mexico (Fig. 1): (a) the Michoacan–Guanajuato volcanic field (MGVF) located north of Uruapan; (b) the Chichinautzin Formation, south of Mexico City.

These volcanic fields consist principally of monogenetic cinder cones that have, in the majority of cases, lavas interstratified with the ashes or volcanic scoria. A characteristic of these monogenetic volcanoes is that they are active only for a short period and then become extinct; in the case of Paricutin in the MGVF, activity started in February 1943 and continued for nine years. All the sampling was carried out in lavas from these monogenetic volcanoes. These volcanoes are related to subduction of the Cocos Plate under Mexico and are calcalkaline in composition (Gonzalez-Huesca 1992).

LOCATION OF SAMPLE SITES

We sampled lavas from 13 different monogenetic volcanoes ranging from 0 to 30 000 years in age, and determined their magnetic directions (declination/inclination) and palaeointensities, as well as studying their rock magnetic properties and determining their chemical compositions. Samples were taken from six monogenetic volcanoes in the Michoacan–Guanajuato volcanic field and seven in the Chichinautzin Formation (Figs 2a and b). We took six to 12 orientated samples at each site, using a petrol-powered diamond drill and a magnetic compass.

The major-element geochemistry of the lavas was determined using an atomic absorption spectrometer (Gonzalez-Huesca 1992). Fig. 3 shows that when K_2O is plotted against silica content the lavas all fall in the calcalkaline field (Peccerillo & Taylor 1976), with basalt–andesites and andesites in the MGVF, and basalts, basalt–andesites and andesites in the Chichinautzin Formation.

The primary criterion used in the selection of sites for this study was the availability of reliable data about the age of the lavas. Unfortunately, very few monogenetic volcanoes have been accurately dated, and not all that have been dated have produced lavas. In Table 1 the ages and locations of the sampled volcanoes are listed, including the ^{14}C data of Bloomfield (1973, 1975), Hasenaka & Carmichael (1985), Martin del Pozzo (1990) and Delgado *et al.* (in press).

As, in general, the charcoal used in the ^{14}C determinations was collected in the soil just below or no more than 20 cm below the layers of ash and lapilli produced by the eruption of the monogenetic volcanoes at distances between 500 and 2000 m from the crater, we can be fairly confident that the ^{14}C ages are a good estimate of the ages of the lavas that erupted from these volcanoes, especially considering the short life-span of these monogenetic volcanoes (less than 15 years).

MAGNETIC MINERALOGY OF THE LAVAS

The identification of the mineral carriers of the magnetic remanence is of vital importance in palaeomagnetic studies, especially in the interpretation of the results, because it can provide direct evidence as to the origin of the magnetization

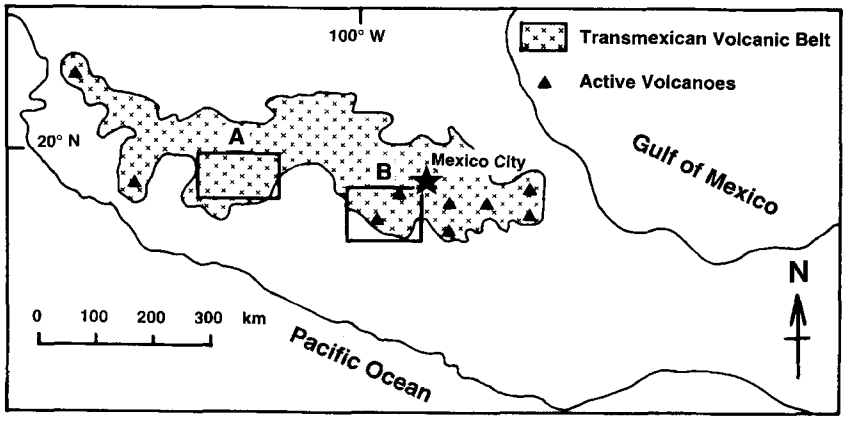


Figure 1. Tectonic map of Mexico showing the general location of the study areas.

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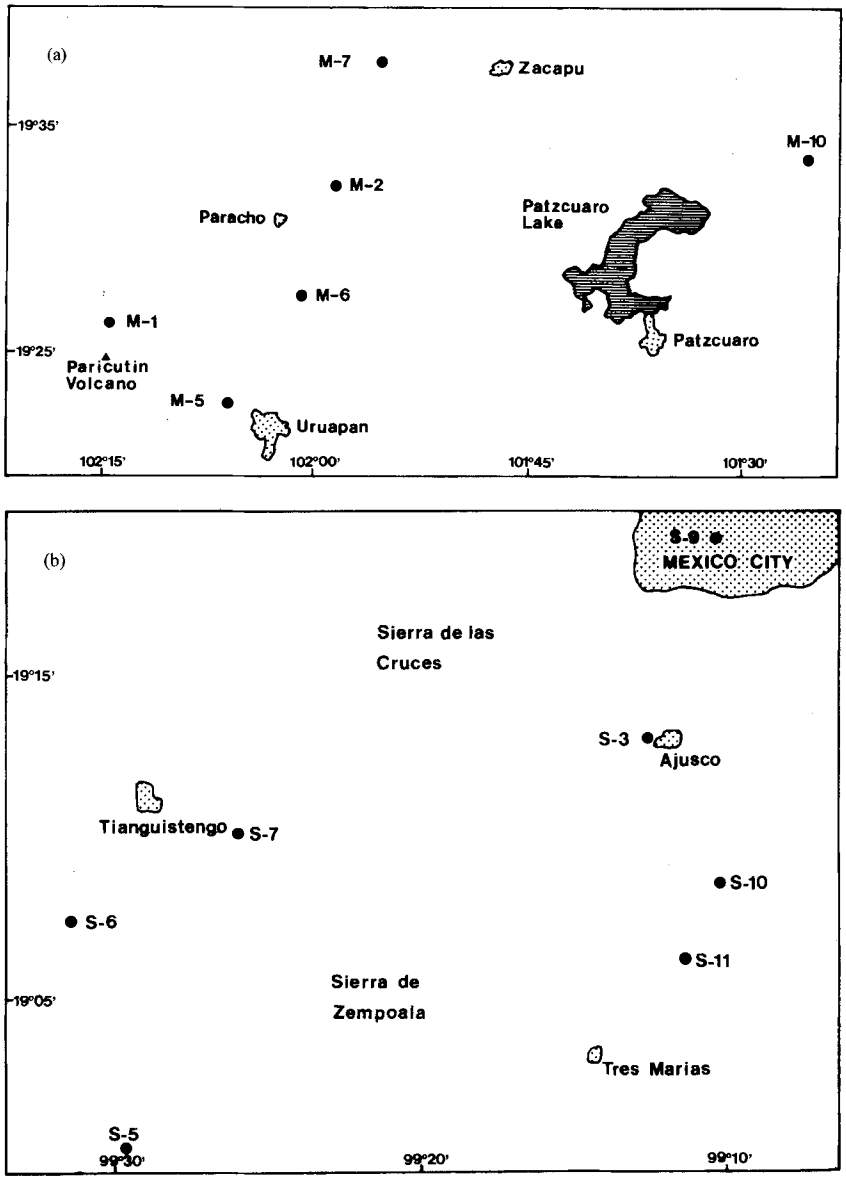


Figure 2. Location maps of the two study areas showing positions of sample sites: (a) Michoacan–Guanajuato volcanic field; (b) Chichinautzin Formation.

as well as the mechanisms by which this remanence was acquired. The best way to do this is by the determination of a combination of several magnetic properties, because it is dangerous to try to rely on just one. The situation is even better when these rock magnetic experiments are combined with some direct observations of the minerals with the reflected light microscope.

Reflected light microscopy

We examined at least one polished surface from each lava flow using reflected light microscopy with oil immersion. The analysis of the samples was made using the optical properties of the magnetic minerals as well as their textural arrangements of oxidation–exsolution, together with the oxidation state (Haggerty 1976). The results of these observations, using oil immersion objectives of 10×, 40× and 100×, are compiled in Table 2. For a number of sites, more than one sample is listed, giving an indication of the variability of the magnetic mineralogy in those particular flows.

In general, the observations show that in the lavas of the monogenetic volcanoes of the Chichinautzin Formation, as well as the rocks of the Michoacan–Guanajuato volcanic field, the dominant magnetic minerals belong to the titanomagnetite series, and are probably carriers of a primary thermoremanent magnetization. As the majority show deuteric oxidation at high temperature (Fig. 4c), indicating that the magnetization was acquired at the time of cooling of the rock, we may assume that the magnetization dates from the time of emplacement of the lavas. It is interesting to note that whilst the lavas have very similar chemical compositions, especially in the Michoacan–Guanajuato Field, where the majority of the sampled lavas are andesites, their titanomagnetites show a wide range of degrees of deuteric oxidation (Table 2), which reflect differing conditions of emplacement temperature, pressure and oxygen fugacity. On the whole, the more basaltic

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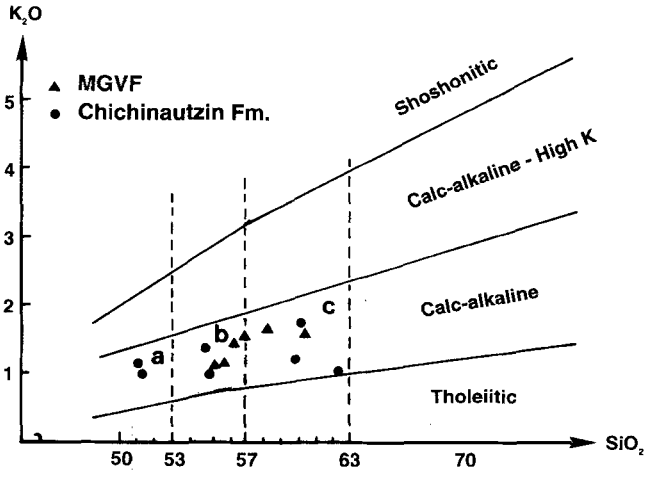


Figure 3. Plot of potassium versus silica (Peccerillo & Taylor 1976), indicating that the compositions (Gonzalez Huesca 1992) of the sampled lavas all fall within the calcalkaline field: (a) basalt; (b) andesite–basalt; (c) andesite.

Table 1. Sample localities and ages.

Site	Volcano	Latitude	Longitude	Age (years)
Chichinautzin Formation				
S-3	Maninal	19.22N	99.21W	18 900 ± 600 BP@
S-5	Tezontle	19.22N	99.47W	21 860 ± 380 BP$
S-6	Tres Cruces	10.12N	99.49W	8440 ± 70 BP$
S-7	Cuautl	19.17N	99.42W	19 530 ± 160 BP$
S-9	Xitle	19.32N	99.18W	1985 ± 55 BP &
S-10	Pelado	19.14N	99.17W	4070 ± 150 BP@
S-11	Cima	19.10N	99.18W	10 160 ± 70 BP@
Michoacan-Guanajuato Volcanic Field				
M-1	Paricutin	19.48N	102.22W	1943 ± 1952 AD*
M-2	El Huanillo	19.67N	101.98W	9180 ± 250 BP*
M-5	El Jabali	19.45	102.11W	3830 ± 150 BP*
M-6	El Metate	19.57N	102.01W	4700 ± 200 BP*
M-7	El Pueblito	19.82N	101.92W	29 000 ± 3300 BP*
M-10	La Mina	19.71N	101.42W	17 170 ± 430 BP*

References: $ Bloomfield (1973, 1975)
* Hasenaka & Carmichael (1985)
@ Martin del Pozzo (1990)
& Delgado *et al.* (in press)

rocks tend to have lower oxidation states (C1–C3) and the more andesitic rocks higher oxidation states (C3–C7).

Particularly interesting are the skeletal textures found in the Tezontle (S-5) lava, because these textures are associated with very quick cooling, and have been found in pillow lavas in the oceanic ridges. The magnetic minerals identified were homogeneous titanomagnetite with an oxidation state C1 and ilmenite R1 (Fig. 4a). In this particular case, the very quick cooling seems to be due to the thinness of the lava flow. In a sample from El Huanillo volcano (M-2), one of the more basaltic lava flows of the MGVF, we found zoned spinel crystals with a corona of Mg-chromite around Ti-magnetite (Fig. 4b).

Rock magnetic properties

Magnetic susceptibility at room temperature

Magnetic susceptibility at room temperature was determined using a Bartington MS2 meter. At the same time, the natural remanent magnetization (NRM) was measured in a Minispin magnetometer. The mean values of each site are shown in Table 3, together with the calculated values of the Konigsberger ratio Q (Königsberger 1938), which is the ratio of the remanent magnetization of the sample to its induced magnetization in the Earth's magnetic field. If the value of Q is less than 1, this normally means that there is a large contribution from multi-domain (MD) grains, but only assuming that the NRM is a thermoremanent magnetization (TRM).

The mean values of the magnetic susceptibility vary from 2.74×10^{-3} to 27.18×10^{-3} for the sampled lavas. Sometimes the NRM and the susceptibility exhibit large variations in one site. However, there is a general correlation of high NRM intensities with high values of magnetic susceptibility, which can be explained by variations in the content of titanomagnetites per unit volume.

In general, the values of the susceptibility of the lavas of the Chichinautzin Formation are lower than the values from the lavas of the Michoacan–Guanajuato volcanic field. This is due to differences in either the compositions or the concentrations of the magnetic minerals. The calculated Konigsberger ratio Q (Table 3) also shows large variations, between 4.53 and 323.45. Some of the high values of Q are associated with the presence of very strong secondary magnetic components, probably related to lightning strikes (see section on palaeomagnetic directions).

Magnetic susceptibility at low temperatures

Magnetic susceptibility at low temperatures was measured using a computer-interfaced Bartington MS2 meter with a water-cooled probe. Samples were mounted in plasticine together with a thermocouple and cooled to 78K in liquid nitrogen. The susceptibility and temperature were recorded as the sample warmed to room temperature.

The majority of igneous rocks display low-temperature susceptibility behaviour that can be assigned to one of three groups on the basis of the shape of their susceptibility versus temperature curves (Senanayake & McElhinny 1981): group 1 samples show an increase in susceptibility with increasing temperature; group 2 samples present a decrease in susceptibility with increasing temperature; and group 3 samples show a peak in susceptibility at a temperature of approximately 125 K.

In addition to these groups, we found during this study another distinctive type of curve in several samples, which is a combination of groups 1 and 2 (Sherwood 1990; Shaw *et al.* 1991), and these we named group 4, where the samples show an initial decrease in susceptibility, followed by a later increase with increasing temperature.

The results of the behaviour of the susceptibility at low temperatures are included in Table 3, which lists the shape of the susceptibility curve, together with the ratio (RS) of the susceptibility at the temperature of the liquid nitrogen to that at room temperature. Sometimes the samples showed 'combinations' of the curves defined by Senanayake & McElhinny (1981), in which case we used symbols such as $\sim 2/3$ or $\sim 1/3$.

In Fig. 5 some typical examples of the results from lavas of the Transmexican Volcanic Belt are shown. Note especially the form for the samples from sites S-5 and S-7, corresponding to group 4.

The variation of the susceptibility at low temperatures gives information about the composition and grain-size distribution

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Table 2. Description of magnetic mineralogy determined using reflected light microscopy.

Sample	TM oxidation	Ilmenite oxidation	Grain size (μm)	Observations
S-30	not det.	–	1–2	Very few minerals
S-3C	C1 & C5–C6		–	1–4 two populations (a) skeletal textures & (b) simplectic textures
S-5J	C1	R1	5–8	Abundant, aligned skeletal textures
S-6A	C1	–	–	Few, euhedral minerals
S-7F	C3–C4	–	1–20	Min. disseminated, v. small, abundant, maghemitized on edges
S-7A	C1–C2	–	3	Dissemination of min with small veins of haematite
S-9E	C4–C5	R5	100–150	Numerous euhedral grains with trellis textures, large quantity of primary ilmenite
S-10E	C1–C2	–	3–15	V. small subhedral grains, sandwich textures & fractures (?maghemitization)
S-11Q	C5–C7 & C1–C3	–	9–35	2 populations—large quantity of anhedral grains, matrix less oxidised
S-11K	C6–C7	–	3–25	Abundant anhedral magnetic minerals, matrix with red haematite pigment
S-11J	C1–C2	–	–	Many subhedral & anhedral grains with sandwich textures
M-1H	C1-C2	–	25	V. few grains, anhedral & subhedral
M-2A	C1	–	3	V. few minerals subhedral some show maghemitized edges
M-5A	C1	–	3–6	Dissemination of very few subhedral grains
M-6A	C3–C5	–	12–25	large number of grains, sandwich texture
M-7S	C1–C3 & C4–C5	–	3–9 & 50	2 populations-many subhedral grains with sandwich texture
M-10A	C2–C4	R1	3-15	Large quantity of subhedral grains with trellis & sandwich textures

Note: Deuteric oxidation classes for titanomagnetite (TM) and ilmenite after Haggerty (1976).

of the magnetic minerals. The interpretation of the curves is complex and sometimes controversial (Radhakrishnamurty, Likhite & Sahasrabudhe 1977; Senanayake & McElhinny 1981, 1982; Radhakrishnamurty 1985; Sherwood 1990; Shaw *et al.* 1991; Radhakrishnamurty & Likhite 1993; Sherwood *et al.* 1993; Sherwood & Basu Mallik 1996). However, there is general agreement about the behaviour of some samples: titanomagnetites rich in Ti ($0.5 < x < 0.7$) show group 1 curves with low *RS* values (< 0.4); only multidomain (MD) magnetite has a group 3 curve, and single-domain (SD) magnetite shows group 1 curves, with high values of *RS* ($0.7 < RS < 1.0$), though this is not diagnostic of SD magnetite.

For group 2 behaviour the situation is less certain. For *RS* values greater than 1 there are a number of possible interpretations: paramagnetic minerals, with susceptibility that varies with the inverse of absolute temperature (Shaw *et al.* 1991); 'ultrafine superparamagnetic particles' (SP+) (Radhakrishnamurty *et al.* 1977); or a very low-Curie-temperature ($< -150\,^{\circ}C$) mineral, such as Ti-rich titanohaematite (virtually pure ilmenite), which is paramagnetic over most of the measured temperature range. In order for paramagnetic minerals to produce *RS* values in excess of 2, a majority of the susceptibility would need to be paramagnetic. Whilst this is a distinct possibility in sedimentary rocks, it is unlikely in fresh, unweathered lavas. For this reason, we favour Ti-rich titanohaematite as the most likely cause of this type of behaviour in andesitic and basaltic lavas. When the value of *RS* is between 1 and 1.4 there exist three additional possibilities in the literature: cation-deficient magnetite (Radhakrishnamurty *et al.* 1977; Radhakrishnamurty 1985); titanomagnetite poor in Ti, divided with many ilmenite lamellae (titanomagnetite with deuteric oxidation Class C2–C3) (Senanayake & McElhinny 1981, 1982); or titanomagnetite with $x = 0.1$ (Radhakrishnamurty 1985; Radhakrishnamurty & Likhite 1993). Of these, only the last is reproducible using artificial samples. The presence of TM10 will be obvious from the Curie temperature, and the presence of Ti-rich titanohaematite can be detected using the reflected light microscope.

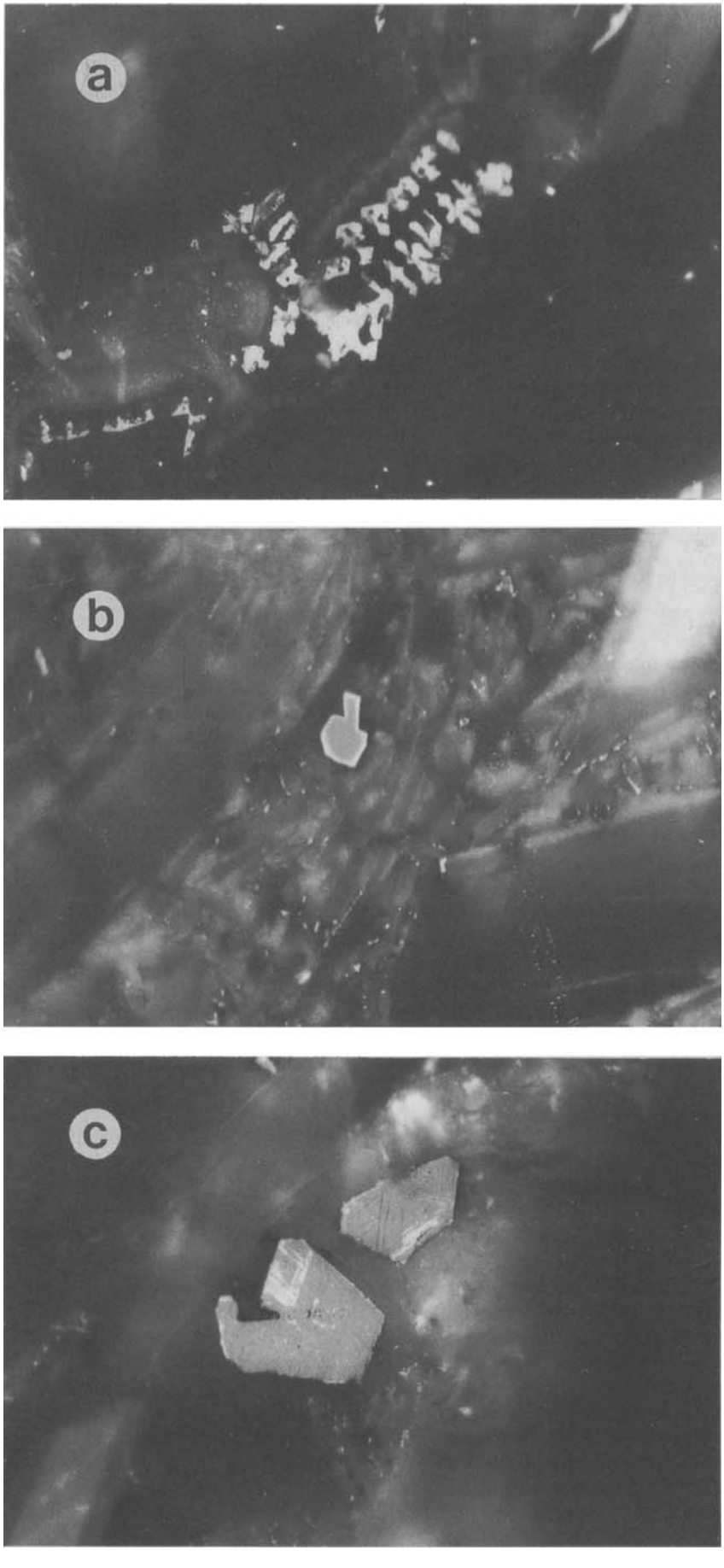
The samples of the MGVF have group 1 and group 3 curves and combinations of these two groups (1/3, indicating a dominance of group 1 with a possible influence of group 3. Three samples, M-2, M-5 and M-7, had values of $RS < 0.4$, suggesting that they contain principally titanomagnetite rich in Ti (without deuteric oxidation), which agrees with the optical observations (Table 2). Samples M-1, M-6 and M-10 have *RS* values of 0.45–0.70, possibly indicating a mix of titanomagnetites poor and rich in Ti.

The samples from the Chichinautzin Formation in general show a different type of low-temperature-susceptibility behaviour. Although the samples from S-6 and S-10 had group 1 curves with *RS* values of 0.25 and 0.50, respectively, characteristic of Ti-rich titanomagnetite, and S-11 showed only a very slight rise in susceptibility with temperature, the majority of the lavas in Chichinautzin have curves associated with groups 2, ~2/3 or 4. These samples show an initial decrease in susceptibility as the temperature rises.

Two samples from site S-9 (Xitle) had very high values of *RS*, suggesting the presence of very low-Curie/Néel-temperature minerals such as Ti-rich titanohaematite. This agrees with the optical observations, which showed large quantities of ilmenite. The magnetic minerals from S-3 have a high oxidation state, with magnetite, ilmenite and sometimes haematite, and in these cases the value of *RS* is 1.13, with group ~2/3 behaviour (group 2 dominant with a slight group 3 influence). Thus, the low-temperature-susceptibility behaviour is probably a combination of SD magnetite, MD magnetite and impure ilmenite.

Optical observations of one of the two group 4 samples, S-5, indicate the presence of skeletal titanomagnetite rich in Ti (C1 class), together with primary ilmenite. This type of behaviour had been reported in a sample of Miocene basalt from Germany rich in primary ilmenite by Sherwood (1990).

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Table 3. Rock magnetic properties.

Site	NRM A/m	K 10^{-3}SI	Q	T_C °C	AR	Low T K grp.	RS	M_S Am^2kg^{-1}	M_{rs} Am^2kg^{-1}	M_{rs}/M_S	H_c mT	H_{cr} mT	H_{cr}/H_c
S-3	1.3	4.1	8.8	620	0.60	~2/3	1.13	0.28	0.11	0.41	44	72	1.6
S-5	3.6	3.0	30.2	295 +60	1.60	4	1.25	0.35	0.10	0.30	14	25	1.8
S-6	9.6	10.0	24.2	280	1.11	1	0.25	0.48	0.09	0.19	10	34	3.4
S-9	3.4	4.6	21.4	610	0.88	2/3	1.60	0.22	0.06	0.28	25	53	2.1
S-10	12.6	2.7	130.9	330 +560	1.36	1	0.50	0.26	0.11	0.43	#21	32	1.5
S-11	3.4	5.4	17.9	625	0.95	1	0.88	0.31	0.10	0.31	34	70	2.1
M-1	43.5	3.8	323.4	290 +490	0.98	1	0.64	0.81	0.36	0.44	44	67	1.5
M-2	26.8	8.4	90.9	550	1.00	1	0.19	0.53	0.23	0.43	22	29	1.3
M-5	33.5	5.0	190.8	440	1.26	1	0.36	0.33	0.16	0.49	29	37	1.3
M-6	13.8	24.2	16.3	410 +560	0.97	~1/3	0.70	1.08	0.14	0.13	11	39	3.5
M-7	15.0	18.1	23.7	540	0.80	1	0.32	1.68	0.35	0.21	15	29	1.9
M-10	64.6	27.2	67.8	315 +520	1.10	~1/3	0.45	1.14	0.21	0.18	15	40	2.7

Note: NRM = Natural remanent magnetization; K = susceptibility at room temperature; Q = Konigsberger ratio (1938); T_c = Curie temperature; AR = alteration ratio during the measurement of the Curie temperature from unitary value at 100°C; low T K grp. = type of varation of the susceptibility at low temperatures; RS = ratio of the susceptibility at −196 °C to the susceptibility at room temperature; M_s,M_{rs} and H_c are all derived from the hysteredsis loop on the MicroMag; H_{cr} is derived from back-field IRM measurements on the MicroMag. # = wasp-waisted hysteresis loop.

Additionally in the same volcanic field (Chichinautzin Formation), Urrutia Fucugauchi, Radhakrishnamurty & Negendank (1984) and Urrutia Fucugauchi, Böhnel & Negendank (1991) reported the same type of behaviour for the San Anton basalt near to Cuernavaca. The curves are interpreted by them as being caused by the presence of SP + grains (Radhakrishnamurty *et al.* 1977). However, from our optical observations we think that this type of curve is probably produced by the combination of titanomagnetites rich in Ti (Group 1) and impure ilmenite (haemenoilmenite).

Curie temperature

Curie temperature was determined using a horizontal balance under computer control. The samples were heated in air, in a magnetic field of approximately 0.5 T to a temperature of 700 °C in 15 min.

Some examples of representative heating–cooling curves are presented in Fig. 6. The Curie temperatures were estimated using the method described by Grommé, Wright & Peck (1969) and are included in Table 3. The Curie temperatures have values from 295 to 625 °C. The low Curie temperatures are associated with states of low deuteric oxidation (C1–C2) with titanomagnetites rich in Ti; the high Curie temperatures, on the other hand, are associated with higher states of deuteric oxidation (C2–C5) with titanomagnetites poor in Ti.

The alteration ratio (*AR*) (Table 3) is equal or near to 1 when the thermomagnetic curves are reversible, and usually different to 1 when the thermomagnetic curves are irreversible.

Some samples (Table 3, Fig. 6) showed two Curie points (S-5, S-10, M-1, M-6 and M-10), indicating the coexistence of two mineral phases, probably titanomagnetite rich in Ti and titanomagnetite poor in Ti. Sometimes the thermomagnetic curves were reversible and sometimes they were irreversible (Fig. 6), probably due to the presence of low-temperature oxidation (maghemitization) in the Ti-rich titanomagnetites.

Samples S-3, S-7, S-9 and S-11 show Curie temperatures higher than the Curie temperature of magnetite (578 °C): this indicates that the magnetic minerals are either members of the ilmenohaematite series or cation-deficient magnetite. In two cases (S-9 and S-11) the optical observations (Table 2) show that the high Curie temperatures are associated with very high oxidation states (C5–C7). In all cases there is a reduction in the magnetization after heating (*AR* ranging from 0.49 to 0.95), and in the case of S-7 there is a reduction in the Curie temperature. This might favour the presence of cation-deficient magnetites, of varying degrees of thermal stability (Ozdemir & Banerjee 1984) in these samples. The hysteresis data (see below) seem to support this.

Generally speaking, the Curie temperatures determined in the samples indicate that the lava flows of the Chichinautzin Formation have higher oxidation states than the lavas of the Michoacan–Guanajuato volcanic field.

Magnetic hysteresis

Magnetic hysteresis was measured using a Princeton Measurements MicroMag 2900 alternating-force magnetometer, using a maximum applied field of 1 T and steps of 5 mT. In all cases there is a noticeable paramagnetic fraction, which causes a linear increase in the induced magnetization after the saturation of the ferromagnetic minerals, which always occurs by about 300 mT, suggesting that haematite is not present in any significant quantity.

Hysteresis parameters, M_{rs}, M_s and H_c (Table 3) have been determined from the hysteresis loops. In order to determine

Figure 4. Photomicrographs of polished sections showing examples of the magnetic mineralogy in these lavas: (a) cruciform skeletal titanomagnetites C1 (total length 35 m) + ilmenite R1, sample S-5J Tezontle, × 100 oil immersion; (b) titanomagnetite with corona of Mg-chromite (size 10 mm), sample M–2 A El Huanillo, × 100 oil immersion; (c) oxidation–exsolution of ilmenite lamellae from titanomagnetite C3–C4: trellis texture, sample S-9E Xitle, × 100 oil immersion.

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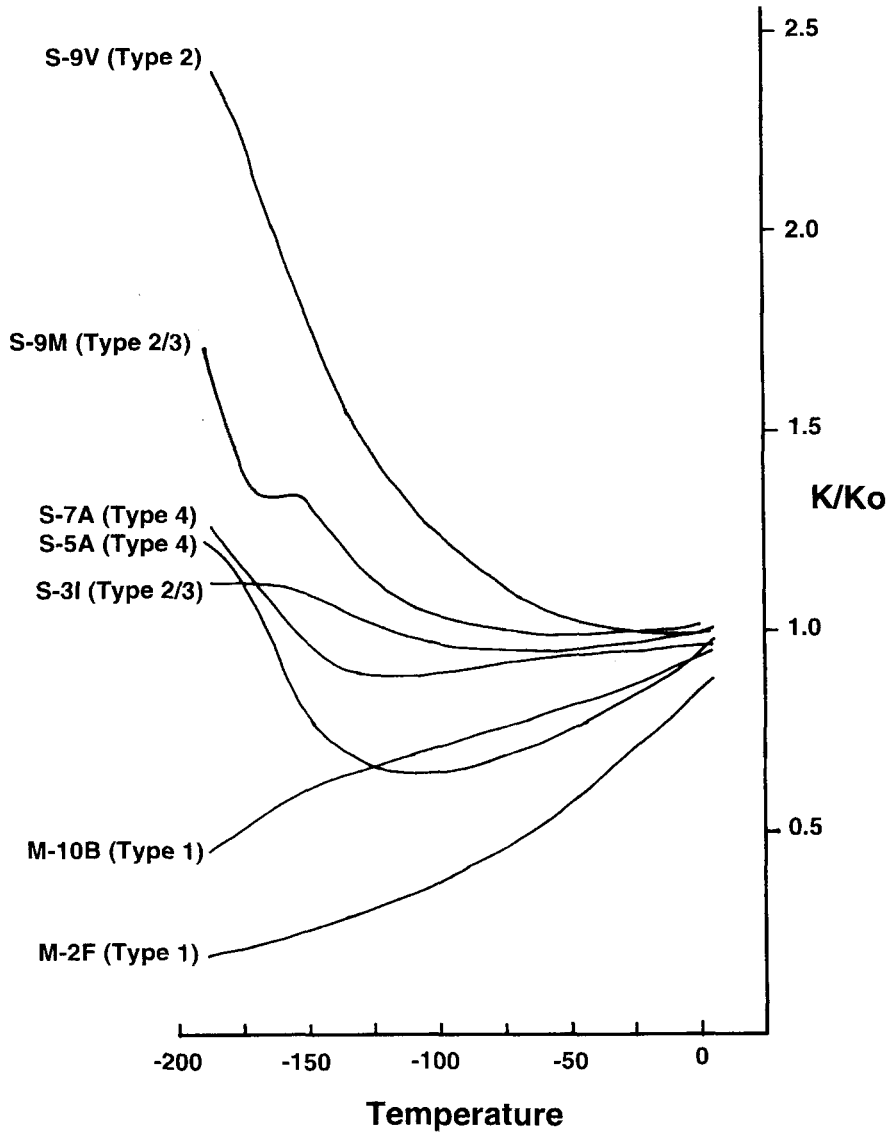


Figure 5. Examples of low-temperature susceptibility behaviour. See text for interpretation and discussion.

the remanence coercivity H_{cr}, samples were given a saturation isothermal remanence (IRM) in a field of 1 T, H_{cr} being the antiparallel field required to reduce the saturation IRM to zero. The ratios M_{rs}/M_s and H_{cr}/H_c have been calculated (Table 3), and plotted against each other (Fig. 7). Most of the samples lie in the pseudo-single domain (PSD) zone of Day, Fuller & Schmidt (1977), but it is more probable that the values represent different combinations of SD and MD grains. In general, the higher values of M_{rs}/M_s are associated with lower values of H_{cr}/H_c, and low values of M_{rs}/M_s are associated with high values of H_{cr}/H_c, so an approximately linear trend can be observed in Fig. 7. We interpret this as a line of mixing with the samples at the top left being those with essentially SD grains and the samples at the bottom right those with predominantly MD grains. The lavas with the higher values of M_{rs}/M_s (>0.4), such as samples S-3, S-10, M-1, M-2 and M-5, probably have mainly SD grains, whereas the samples with the lower values (<0.2), such as S-6, M-6 and M-10, have more MD grains.

The results indicate that the magnetic carriers in these lavas are predominantly titanomagnetites (with variable quantities of Ti), with different forms and states of deuteric oxidation, derived from different conditions during the emplacement of the magmas. The lavas of the Chichinautzin Formation tend to have higher deuteric oxidation states and have different magnetic properties than the lavas of the Michoacan–Guanajuato volcanic field.

In general, the interpretations of the different magnetic properties are in agreement or are complementary, and they fit with our observations on polished sections using the reflected light microscope. However, there are some sites that do show discrepancies in the magnetic properties and, hence, result in contradictory interpretations as to the magnetic mineralogy present in the lavas. We believe this is the result of the occasional use in the various rock magnetic experiments of different samples from the same lava flow but from different positions in the flow. This shows that even in very thin lava flows (and the majority of the lavas sampled in the two volcanic

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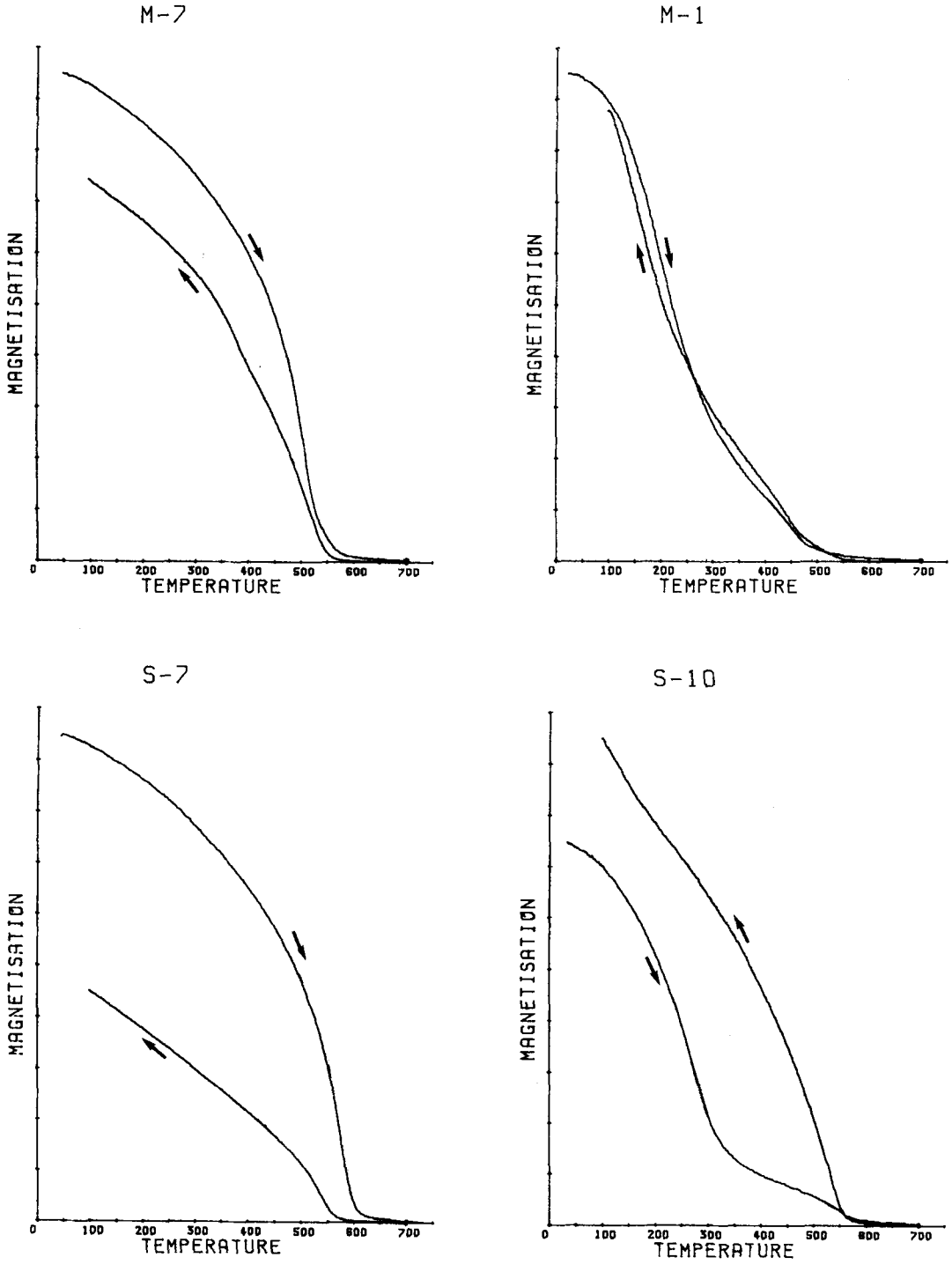


Figure 6. Examples of thermomagnetic curves. M-7: sample with Ti-poor titanomagnetite, $T_c = 540\,^{\circ}C$; M-1: two coexistent phases of titanomagnetite, $T_c = 290$ and 490 °C; S-7: probably cation-deficient magnetite, $T_c = 605\,^{\circ}C$; S-10: Ti-rich titanomagnetite, probably affected by low-temperature oxidation, hence irreversible, $T_c = 330$ and 560 °C.

fields were less than 2 m thick), the variation in the magnetic mineralogy can be significant.

PALAEOMAGNETIC DIRECTIONS

Palaeomagnetic directions have been determined from all 13 sites. Standard 25 mm diameter orientated cores were used for these measurements. All NRM measurements were performed using a Molspin spinner magnetometer. In order to determine the stability of the magnetization and its vectorial structure, samples were subjected to step-wise demagnetization using either alternating-field demagnetization with eight or more steps and a maximum field of at least 80 mT, or thermal demagnetization in 10 steps to a maximum temperature of 670 °C.

At least five samples from each site were measured. In the

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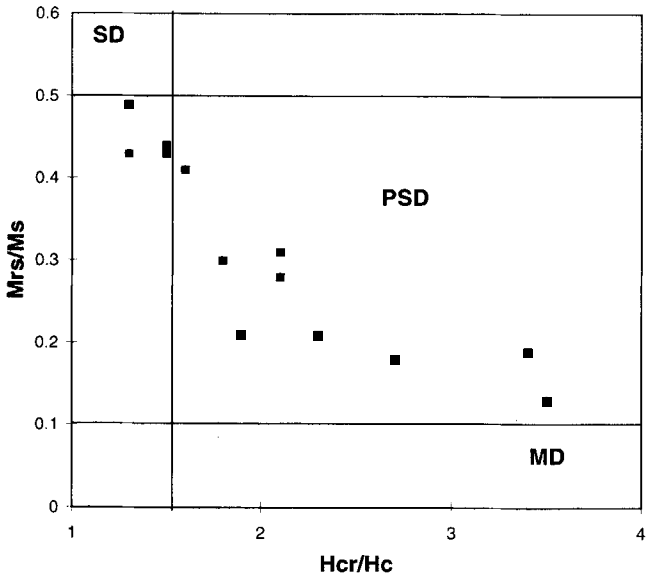


Figure 7. Hysteresis parameters (Table 3) plotted on the diagram of Day *et al.* (1977); see text for interpretation.

majority of cases alternating-field demagnetization was used. The stepwise data were plotted on orthogonal plots (Zijderveld 1967), examples of which are shown in Fig. 8. The majority of samples possess a single-component NRM (Fig. 8a). There are a significant number of samples in which the magnetization is complex (Fig. 8b), with a second strong, often randomly orientated, component of magnetization overprinting the primary NRM; sometimes the original direction is completely destroyed by the later magnetization. One possible explanation is that the overprint is an IRM induced by a lightning strike. Many of the sites with these overprints have very high Königsberger (1938) ratios (Table 3), supporting the inference that the NRM is an IRM. Another possible cause for the secondary component is low-temperature oxidation occurring in the rock after its formation, with the new cation-deficient titanomagnetite recording a chemical remanent magnetization (CRM) parallel to the geomagnetic field at that time. It is possible, for example, that this occurred at site M-2, as maghemite was observed under the microscope (Table 2). Sites S-5 and S-7 were resampled in a later field season in a slightly different place, as the first sets of samples were completely overprinted with a random strong magnetization, almost certainly a lightning strike. The new samples displayed a single component of magnetization.

The palaeomagnetic direction was determined either by taking a stable endpoint, or by principal component analysis, depending on the quality of the data. Site mean directions were calculated using Fisher (1953) statistics, along with their virtual geomagnetic poles (VGPs) (Table 4). Single-component magnetizations were recorded in 10 out of the 13 sites, although a further set of samples from S-10, acquired in order to improve the directional grouping, all showed a shallow northwesterly overprint and no stable endpoint. In most cases the directional grouping is good (high k and low a_{95}), and there were very few anomalous samples that had to be rejected.

The samples from the Michoacan–Guanajuato volcanic field were less well behaved palaeomagnetically than those of the Chichinautzin Formation. As mentioned above, some of the Chichinautzin sites had to be sampled a second time in order to obtain good palaeomagnetic directions, but this was not possible in the case of the MGVF.

The use of a magnetic compass may have introduced additional orientation errors that could have been avoided by the use of a sun-compass. However, unless there is a systematic error due to a local deviation of the magnetic field, random sampling in the field should reduce any systematic effect of the rocks on the compass. Truly random sampling was usually difficult, given the nature of these lava flows; they could only be sampled where the centre of the flow was exposed, as the tops usually consisted of rubble or blocks. However, given that random sampling was attempted, the major effect of using a magnetic compass should be an increase in the scatter of the results, which is not observed. Clearly, the most strongly magnetized rocks will have the greatest effect on the compass; however, as these rocks seem to have been struck by lightning, orientation errors are the least of the problems in these samples.

The direction obtained from site M-1, Paricutin, a volcano that was active between 1943 and 1953, is: Dec. = 10.7°, Inc. = 37.8°, $a_{95} = 4.4°$. The direction recorded at the magnetic observatory in Teoloyucan at that time was: Dec. = 9.5°, Inc. = 47.1° (Campos Enríquez, Campos Enríquez & Urrutia Fucugauchi 1991). A similar inclination error was observed in a further study of the Paricutin lava (Alva-Valdivia & Urrutia Fucugauchi 1996). The difference in the inclination is difficult to explain; it is possibly due to a local magnetic anomaly.

PALAEOINTENSITY EXPERIMENTS

One of the aims of this study was to obtain a complete record of palaeosecular variation in Central Mexico, and palaeointensity data form an essential part of this. For a successful palaeointensity experiment it is necessary to have a good sample. Views vary as to what constitutes a 'good' palaeointensity sample, but there are certain prerequisites. First, the sample must have a TRM as its NRM: this means that samples with secondary overprints of IRM (e.g. lightning strikes) or CRM (e.g. widespread maghemitization) are unsuitable, unless there is some way of isolating the primary TRM. Second, thermal alteration, and the formation of new minerals, during the palaeointensity experiment are highly undesirable, and must be detected. In some instances it may be possible to compensate for thermal alteration (Rolph & Shaw 1985), but often samples have to be discarded once thermal alteration sets in.

There are two widely used methods of palaeointensity determination, both of which use the fact that TRM varies linearly with applied field in weak fields such as the geomagnetic field (Nagata 1943). Both the Thellier & Thellier (1959) and the Shaw (1974) techniques were used in this study (Table 5). Thellier experiments, using the Coe (1967a,b) modification, were carried out at the University of Münster in Germany using a Digico spinner magnetometer and an oven designed for Thellier experiments (Schnepp 1991). Shaw experiments, with the Rolph & Shaw (1985) correction, were carried out at the University of Liverpool using a purpose-built automated cryogenic magnetometer system (Shaw, Share & Rogers 1984).

40 samples were measured in Thellier palaeointensity experiments. Of these 32 gave acceptable results (Tables 5 and 6). Nine or 10 heating steps were used, the exact temperatures being adjusted to the blocking-temperature spectra of the samples to give a fairly even spread of points. The laboratory

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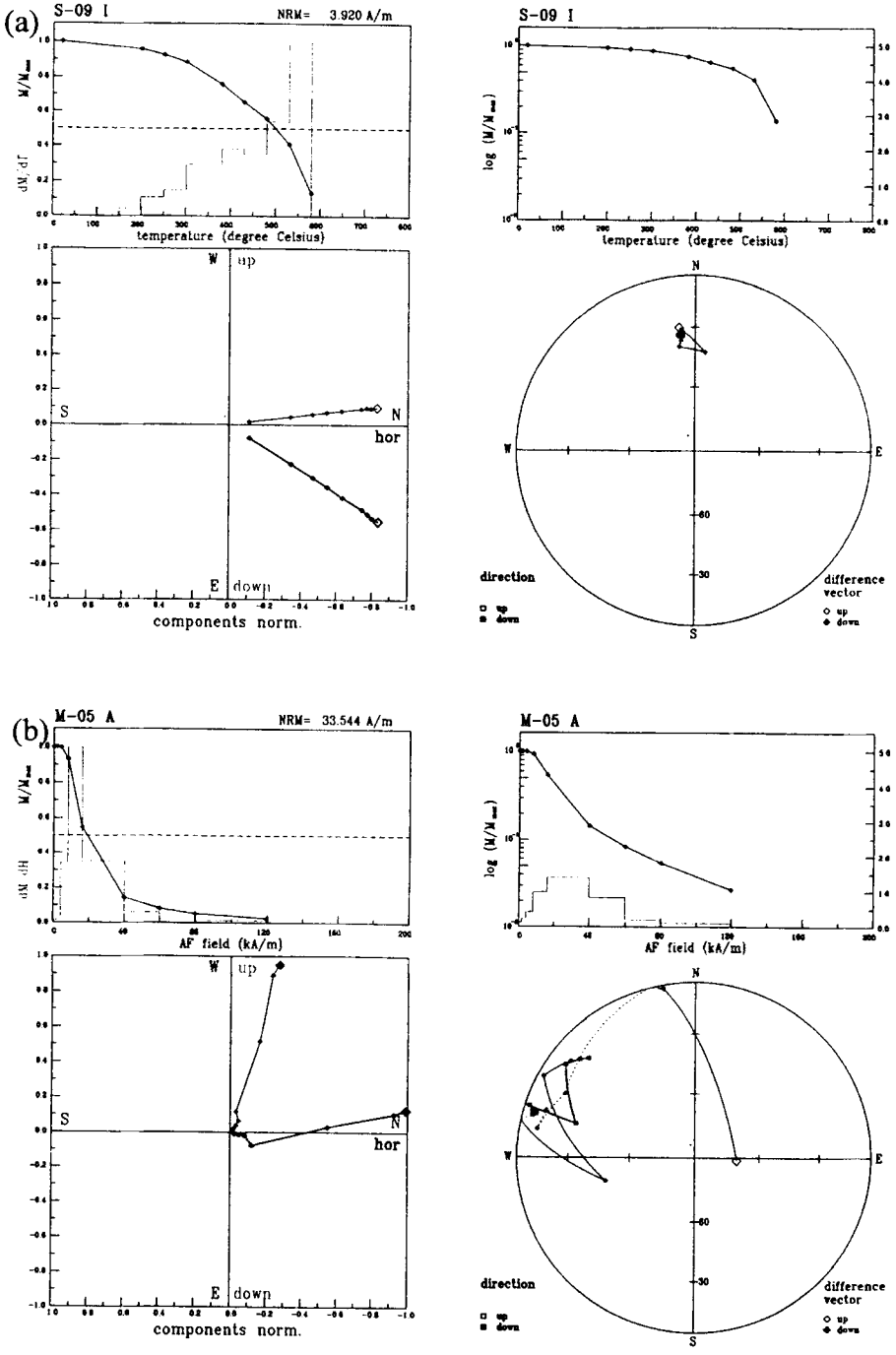


Figure 8. Examples of palaeomagnetic behaviour during stepwise demagnetization, including orthogonal plots (Zijderveld 1967). (a) Thermal demagnetization of sample from S-9 Xitle, showing a single component of magnetization; (b) alternating-field demagnetization of sample from M-5 El Jabali, showing strong secondary magnetization (possibly caused by a lightning strike) overprinting the primary direction.

field was set at a value close to the expected value of the ancient geomagnetic field intensity. During the experiment two or three repeat partial TRMs (pTRMs) were made to check for thermal alteration (Coe *et al.* 1978). After each double-heating step, the susceptibility was measured as a further control on thermal alteration. The experiment was abandoned once the NRM was fully removed or the pTRM checks failed.

The data were plotted (Fig. 9) as NRM versus TRM (Nagata, Arai & Momose 1963), along with the pTRM checks. A line of best fit was drawn and the palaeointensity calculated. In addition, a range of parameters, devised by Coe *et al.* (1978), were calculated (Table 6) for each Thellier experiment: the fraction factor (f), the fraction of the NRM used in the palaeointensity estimate; the gap factor (g), the evenness of the spacing of the points (this may vary between 0 and 1, the higher the better); the coefficient of correlation (CC); and the quality factor (q), a combination of f, g and the error in the palaeointensity (again, the higher the better).

Samples were accepted if they met the following criteria: the linear regression used at least four points; the correlation

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Table 4. Site mean palaeomagnetic directions and VGPs.

Site	n/N	Dec.	Inc.	k	a_{95}	VGP Lat	VGP Long	dp	dm
Lavas with single-component magnetization									
S-3	5/5	359.1	33.7	174.5	5.8	88.8	128.5	3.8	6.6
S-5	7/7	353.3	64.4	318.2	3.4	62.2	250.5	4.4	5.4
S-6	6/6	337.4	55.9	462.7	3.1	63.7	216.3	3.2	4.4
S-7	6/6	342.6	16.6	255.1	4.2	70.0	140.6	2.2	4.3
S-9	9/9	350.1	35.4	662.8	2.0	80.7	173.9	1.3	2.3
S-10	7/9	351.8	12.4	59.8	7.9	74.9	113.7	4.1	8.0
S-11	7/7	354.6	40.8	138.8	5.1	83.4	211.9	3.8	6.2
M-1	6/6	10.7	37.8	237.5	4.4	79.8	336.2	3.1	5.2
M-6	5/5	82.0	41.5	301.1	4.4	14.0	327.3	3.3	5.4
M-10	6/6	339.7	58.2	212.8	4.6	64.0	220.5	5.0	6.8
Lavas with multi-component magnetization									
M-2	3/7	42.2	10.7	540.0	5.3	46.6	1.4	2.7	5.4
M-5	3/6	20.3	44.9	41.6	19.4	70.1	323.5	15.5	24.5
M-7	5/6	3.6	39.9	227.0	5.1	85.6	306.9	3.7	6.1

Note: n/N = number of samples used in Fisher statistics/total number of samples measured; Dec, Inc = declination and inclination of the site mean direction; k, a_{95} = statistical parameters (Fisher 1953); VGP Lat, VGP Long = latitude and longitude of the mean virtual geomagnetic pole (VGP) for the site; dp, dm = errors in the VGP.

coefficient was greater than 0.90; the fraction factor was greater than 0.1; and the quality factor was greater than 0.25. Amongst the samples rejected on these criteria were all those with a secondary overprint. The secondary magnetization could not be removed before thermal alteration occurred in the samples.

The Shaw technique uses a single heating to above the Curie temperature of the rock in order to impart a TRM. This means that thermal alteration is more of a problem than in the Thellier experiment, where several heating steps have often been measured before thermal alteration sets in. Rolph & Shaw (1985) proposed a method of compensating for thermal alteration that looked at the ratios at each demagnetization step of the two anhysteretic remanences (ARMs) given to the samples before and after the imposition of the laboratory TRM in order to check for thermal alteration, and multiplied the TRM by the ratio of ARM1 to ARM2.

A total of 41 Shaw experiments, using the experimental technique described by Sherwood *et al.* (1993) have been made on these lavas (Table 5, Fig. 10), and using the Rolph & Shaw (1985) correction, palaeointensity values have been calculated from most of them, including many where the ARM1–ARM2 graph was non-linear. It is less easy to define rejection criteria when using a correction method that produces an answer from most samples, but we used only the part of the coercivity spectrum that represented the primary TRM, rejecting points that represented a secondary overprint.

When looking at the palaeointensity results (Table 5) it is clear that in many cases there is significant scatter within single lava flows. There is no systematic difference between the Thellier and Shaw results, though the scatter of the Shaw data tends to be higher. This is probably due to the lack of stringent quality control criteria in the Shaw experiments. A number of results have been rejected (Table 5), and were not used in the calculation of the site mean palaeointensity values (Table 7). These were results that were significantly different from the other values obtained for that site. In some cases abnormally high values were obtained using both Shaw (M-7D) and Thellier (S-9K) methods. This type of behaviour has been

Table 5 Results of palaeointensity experiments using Thellier & Shaw methods.

Sample	Volcano	Palaeointensity (μT)	Method	A/R
S-3Ha2		34.86 ± 3.39	Thell	A
S-3Ia		31.29 ± 3.25	Thell	A
S-3Ib		21.87 ± 1.35	Shaw	A
S-3Ka	Maninal	37.93 ± 9.85	Thell	A
S-3Fa		30.51 ± 1.68	Thell	A
S-3Ja		26.54 ± 3.32	Thell	A
S-3Jb		24.71 ± 0.27	Shaw	A
S-5Aa2		no result	Thell	R
S-5Ab2		25.02 ± 1.35	Shaw	A
S-5E		no result	Thell	R
S-5H	Tezontle	no result	Thell	R
S-5L	(Tenago)	no result	Thell	R
S-5Ca		49.90 ± 0.81	Shaw	R
S-5A		14.52 ± 1.37	Shaw	A
S-5Gb		19.41 ± 0.34	Shaw	A
S-6Bb		26.75 ± 0.62	Shaw	A
S-6Cb		56.13 ± 0.61	Shaw	A
S-6Db	Tres Cruces	no result	Shaw	R
S-6Eb		62.01 ± 1.67	Shaw	A
S-6Gb		33.01 ± 0.91	Shaw	A
S-7Ba		23.05 ± 0.52	Shaw	A
S-7Cb		16.79 ± 0.38	Shaw	A
S-7Ca		24.42 ± 0.70	Shaw	A
S-7Fa	Cuautl	33.57 ± 0.21	Shaw	A
S-7Ic		19.62 ± 1.30	Shaw	A
S-7Ab		30.75 ± 0.40	Shaw	A
S-9B		76.64 ± 2.25	Thell	A
S-9C		54.23 ± 3.23	Thell	A
S-9E		59.33 ± 5.39	Thell	A
S-9H	Xitle	79.40 ± 2.06	Thell	A
S-9I		70.03 ± 1.22	Thell	A
S-9K		119.80 ± 14.53	Thell	R
S-9GH		61.35 ± 2.71	Shaw	A
S-9Sin		37.19 ± 1.06	Shaw	R
S-10A		83.51 ± 4.78	Thell	A
S-10E		68.33 ± 3.23	Thell	A
S-10B		no result	Thell	R
S-10Aa	El Pelado	31.83 ± 1.06	Shaw	A
S-10Bb		32.90 ± 0.28	Shaw	A
S-11Ab		29.27 ± 0.89	Thell	A
S-11Aa		36.50 ± 0.20	Shaw	A
S-11Ba		26.10 ± 1.87	Thell	A
S-11Ca		22.41 ± 1.06	Thell	A
S-11Ka	Cima	23.75 ± 1.85	Thell	A
S-11Kb		28.91 ± 0.39	Shaw	A
S-11La		19.29 ± 4.84	Thell	A
S-11Oa		no result	Thell	R
S-11Ea		29.72 ± 0.05	Shaw	A
M-1A		56.07 ± 0.62	Thell	A
M-1Ca		67.62 ± 1.60	Thell	A
M-1N		45.67 ± 0.46	Shaw	A
M-1Ta	Paricutin	40.48 ± 0.50	Shaw	A
M-1Y1		36.10 ± 0.35	Shaw	A
M-2Ba		80.62 ± 4.82	Thell	R
M-2Ca		63.35 ± 0.69	Thell	A
M-2Fa	El Jabali	49.89 ± 2.02	Thell	A
M-2CSin		36.15 ± 1.11	Shaw	A
M-2Ga		47.67 ± 1.04	Shaw	A
M-21		48.91 ± 0.97	Shaw	A

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Table 5. (*Continued.*)

Sample	Volcano	Palaeointensity (μT)	Method	A/R
M-51		30.35 ± 0.83	Shaw	A
M-53R	El Jabali	33.47 ± 0.38	Shaw	A
M-52		22.80 ± 0.55	Shaw	A
M-5Bb		60.57 ± 1.17	Shaw	R
M-6A		54.47 ± 11.72	Thell	A
M-6Sin		56.72 ± 5.29	Thell	A
M-6C	El Metate	53.68 ± 2.79	Thell	A
M-6F		53.08 ± 2.19	Thell	A
M-61Sin		44.77 ± 0.41	Shaw	A
M-6P		54.43 ± 3.74	Shaw	A
M-7Ba		no result	Thell	R
M-7Ea		no result	Thell	R
M-7Aa	El Pueblito	19.23 ± 0.19	Shaw	A
M-7Ia		38.80 ± 1.57	Shaw	A
M-7C		14.65 ± 0.89	Shaw	A
M-7D		118.49 ± 9.93	Shaw	R
M-10A		55.12 ± 5.43	Thell	A
M-10Ca		46.14 ± 2.86	Thell	A
M-10E	La Mina	42.29 ± 3.79	Thell	A
M-10F		97.03 ± 1.15	Thell	R
M-10G		37.68 ± 2.17	Thell	A
M-10B		38.53 ± 0.27	Shaw	A

Note: Thell = Thellier with modification of Coe (1967); Shaw = Shaw with modification of Rolph & Shaw (1985); A = accepted; R = rejected.

Table 6. Additional data (Coe *et al.* 1978) from Thellier experiments.

Sample	$T_i..T_f$	n/N	CC	f	g	q
S-3Ha2	100–451	6/10	−0.990	0.914	0.754	7.089
S-3Ia	211–451	5/10	−0.993	0.396	0.709	2.708
S-3Ka	110–352	5/10	−0.930	0.138	0.738	0.391
S-3Fa	20–401	6/9	−0.997	0.519	0.725	6.821
S-3Ja1	205–550	5/9	−0.991	0.383	0.683	2.091
S-9B	20–401	7/10	−0.995	0.228	0.780	6.043
S-9C	20–430	6/9	−0.988	0.187	0.731	2.292
S-9E	251–451	4/10	−0.987	0.246	0.589	1.592
S-9H	20–451	8/10	−0.995	0.391	0.779	11.713
S-9I	20–530	8/9	−0.998	0.589	0.819	27.817
S-9K	211–401	4/10	−0.951	0.127	0.480	0.503
S-10A	20–430	6/9	−0.982	0.349	0.549	3.346
S-10E	20–301	4/9	−0.995	0.502	0.650	6.899
S-11Ab	199–580	8/9	−0.999	0.794	0.558	14.643
S-11Ba	20–430	6/9	−0.996	0.114	0.739	1.174
S-11Ca	20–601	10/10	−0.999	0.862	0.811	14.825
S-11Ka	310–601	6/10	−0.998	0.769	0.751	7.436
S-11La	252–451	5/10	−0.991	0.432	0.609	1.050
M-1A	20–401	5/9	−1.000	0.879	0.669	52.894
M-1Ca	204–352	4/10	−0.999	0.562	0.610	14.464
M-2Ba	204–352	4/10	−0.992	0.379	0.609	3.857
M-2Ca	110–352	5/10	−1.000	0.632	0.700	40.850
M-2Fa	20–252	4/10	−0.998	0.307	0.336	2.549
M-6A	204–451	6/10	−0.862	0.107	0.766	0.383
M-6Sin	110–451	7/10	−0.967	0.101	0.748	0.813
M-6C	199–530	7/9	−0.989	0.402	0.622	4.814
M-6F	199–530	7/9	−0.993	0.332	0.621	4.982
M-10A	199–381	4/9	−0.983	0.173	0.639	1.125
M-10Ca	20–381	5/9	−0.992	0.549	0.676	5.996
M-10E	251–501	5/10	−0.990	0.400	0.738	3.297
M-10F	20–352	6/10	−0.999	0.533	0.767	34.431
M-10G	251–501	5/10	−0.997	0.485	0.711	5.984

Note: $T_i..T_f$ = Range of temperature used; n/N = number of points accepted for the line of regression/total number of points; CC = coefficient of correlation; f = fraction factor; g = gap factor; q = quality factor (Coe *et al.* 1978).

detected before (Sherwood 1991) in lavas, and though in a few cases it may be caused by incomplete removal of a secondary overprint, often it is more difficult to explain. It has been suggested that either these samples have a thermochemical remanence, rather than a pure TRM, or some undetectable thermal alteration has occurred during the experiment.

Site M-1, Paricutin, again provides some quality control on the palaeointensity results: the site mean palaeointensity is 49.2 ± 12.7 μT, the observed field intensity at the time of eruption was 45 μT (Campos *et al.* 1991). The palaeointensity mean is raised by the Thellier results, which seem to be far higher than the actual field; on the other hand, the Shaw results are equal to or slightly lower than the actual field.

A number of Thellier palaeointensity experiments on the Xitle lava (S-9) were carried out by Nagata, Kobayashi & Schwarz (1965). Their mean value of 57.2 ± 5.4 μT is slightly lower than ours (66.8 ± 10.1 μT). We are now undertaking a very detailed palaeomagnetic and rock magnetic study of this flow, and have already obtained nearly 100 palaeointensity estimates from it (Böhnel *et al.* 1995; Sherwood *et al.* 1995). It seems from this new work that the mean value obtained from just six samples is a reasonable estimate of the palaeointensity of this flow.

The site mean palaeointensities and the associated virtual dipole moments (VDM) (Smith 1967) are listed in Table 7. In three cases, the error is excessive (S-6, S-10 and M-7), and closer inspection shows the palaeointensity estimates from these lavas clustering around two very different values. It is impossible to decide which, if any, reflects the true palaeointensity, so these results must be considered dubious. It seems that the scatter is lower in those lavas with a single component of magnetization, a high (> 450 °C) Curie temperature, and which exhibit high-temperature deuteric oxidation (C3-C7), although these criteria do not guarantee a good result. As the VDM uses the inclination to reduce the palaeointensity value to the equivalent geocentric dipole, its accuracy will depend on those of both the palaeointensity determination and the palaeomagnetic direction.

SECULAR VARIATION IN CENTRAL MEXICO OVER THE LAST 30 000 YEARS

Fig. 11 shows our palaeosecular variation curves for the last 30 000 years in Central Mexico, based upon the 13 site mean directional and palaeointensity results described above. The inclination varies between 12.7 and 64.4 °, and the declination between 337.4 and 82.9 °, indicating that quite a high-amplitude secular variation has occurred in the last 30 000 years. The swings in declination are the most marked, in particular the eastward swing at about 5000 years recorded by M-6, El Metate. This is a well-constrained direction, with an a_{95} of 4.4 ° and a VGP latitude of 14 °N, and there are no obvious reasons to doubt its accuracy. The northeasterly declination of M-2 is less well constrained, a stable direction only being determined from three out of seven cores; however, with a VGP latitude of 46.6 °N it is not far off the normal range of directions.

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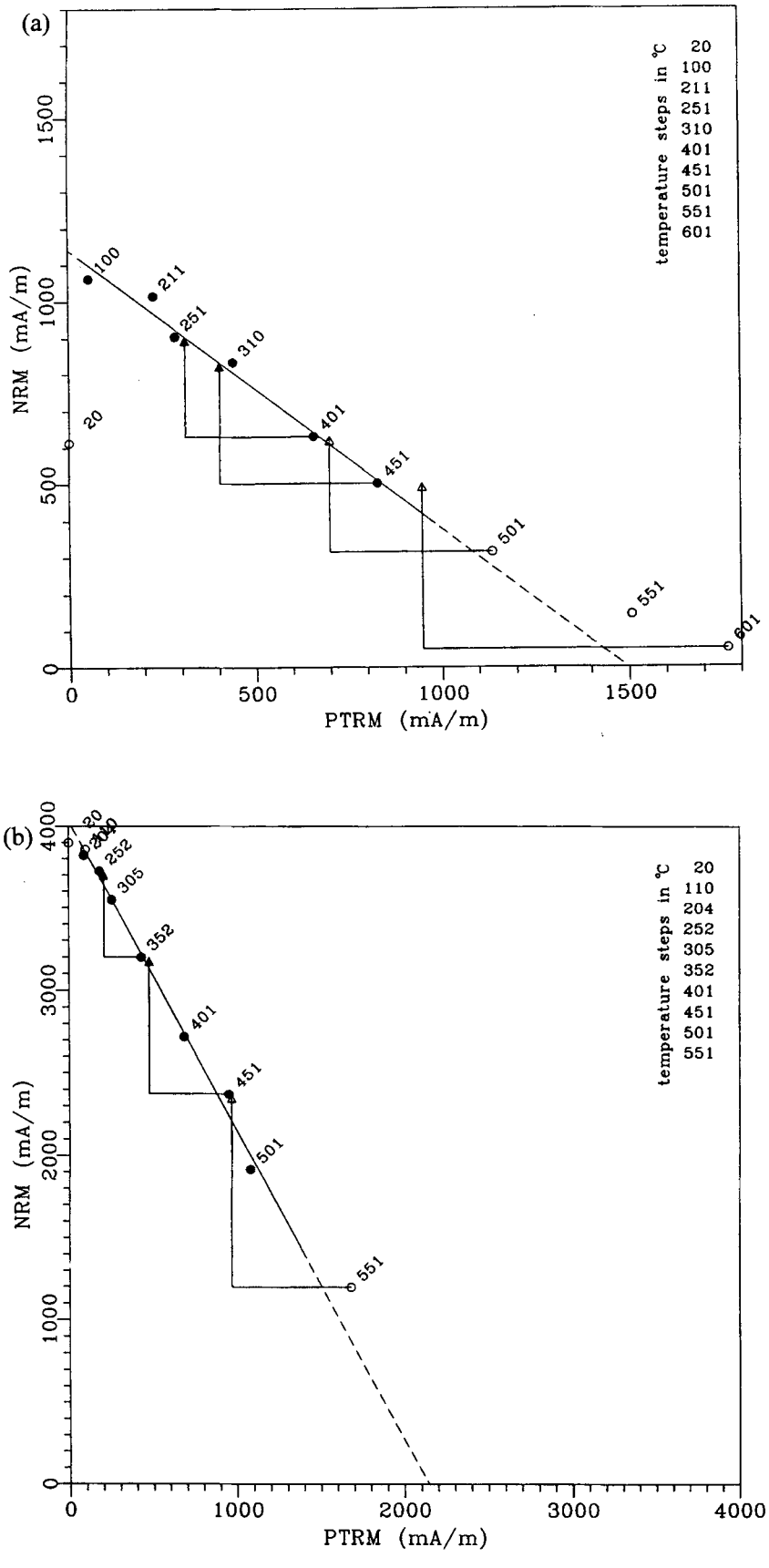


Figure 9. Examples of successful modified Thellier (Coe 1967a,b) palaeointensity experiments. (a) S-3Ha2 Maninal, palaeointensity = 34.86 ± 3.39 μT; (b) S-9H Xitle, palaeointensity = 79.40 ± 2.06 μT. Triangles represent pTRM checks.

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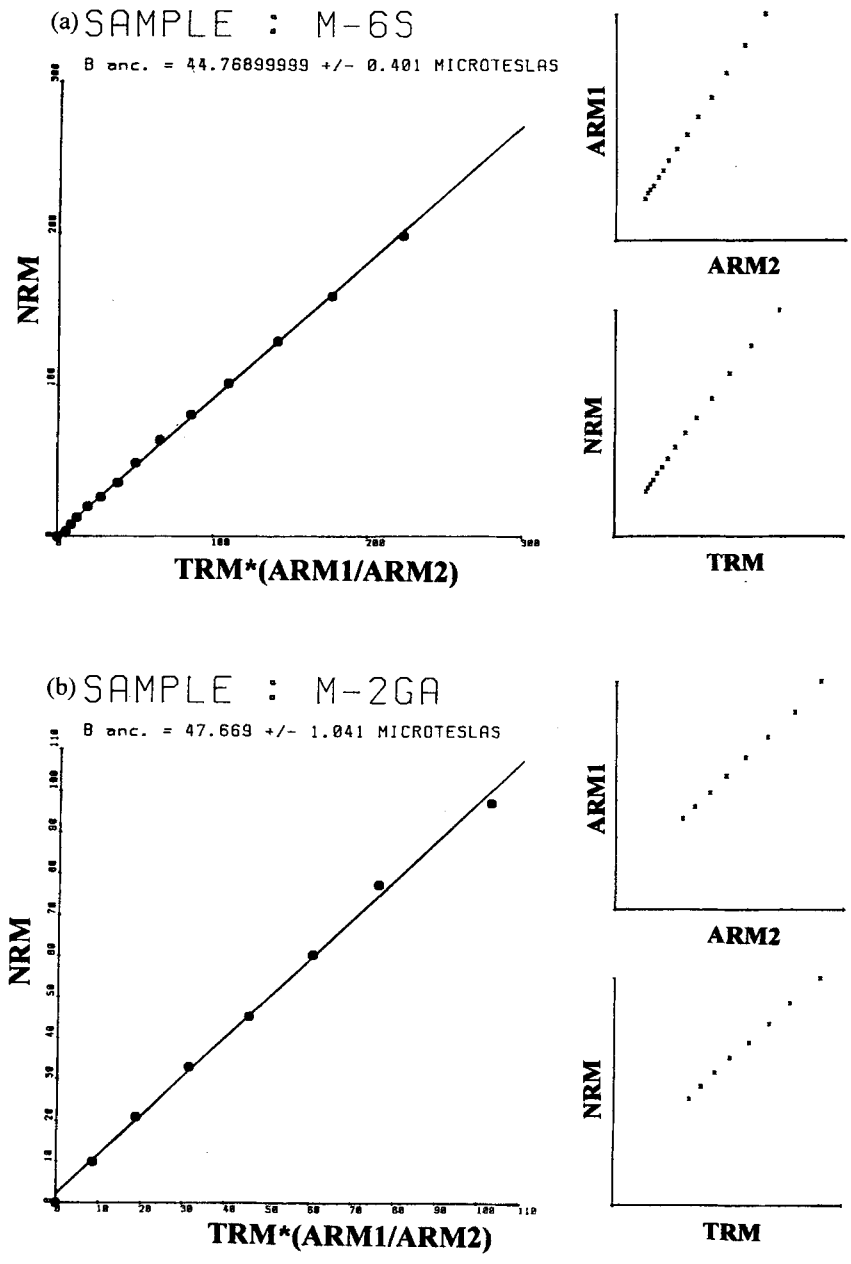


Figure 10. Examples of successful Shaw palaeointensity experiments plotting using the correction of Rolph & Shaw (1985). (a) M-6S El Metate, palaeointensity = 44.77 ± 0.40 μT; (b) M-2GA El Huanillo, palaeointensity = 47.67 ± 1.04 μT.

Table 7. Site mean palaeointensities and virtual dipole moments.

Site	Age (years)	Mean palaeointensity (μT)	VDM (10^{22} Am^2)
S-3	18 900 ± 600	29.7 ± 5.7	6.7 ± 1.3
S-5	21 860 ± 380	19.4 ± 5.3	3.1 ± 0.4
S-6	8440 ± 70	44.5 ± 17.2	8.0 ± 3.1
S-7	19 530 ± 160	24.7 ± 6.4	6.2 ± 1.6
S-9	1985 ± 55	66.8 ± 10.1	14.9 ± 2.3
S-10	4070 ± 150	54.1 ± 25.9	13.7 ± 6.6
S-11	10 410 ± 80	27.0 ± 5.3	5.8 ± 1.1
M-1	1943–1952 AD	49.2 ± 12.7	10.8 ± 2.8
M-2	9410 ± 230	49.2 ± 9.7	12.6 ± 2.5
M-5	3830 ± 150	28.9 ± 5.5	5.9 ± 1.1
M-6	4700 ± 200	52.9 ± 4.2	11.2 ± 0.9
M-7	29 000 ± 3300	24.2 ± 12.8	5.2 ± 2.8
M-10	17 170 ± 430	44.0 ± 7.1	7.7 ± 1.2

It is interesting to compare our directional data with directions obtained from lake sediments in Central Mexico: Liddicoat *et al.* (1979) reported no anomalous directions at around 5000 years in sediments from Tlapacoya in the Valley of Mexico; however, Ortega-Guerrero (1992) has recently completed a palaeomagnetic study of lake sediments near Chalco in the Valley of Mexico. Results from one of her cores (Urrutia Fucugauchi *et al.* 1995) indicate a strong easterly swing in declination at about 5000 years BP, and another smaller one at about 14 000 years BP (a period for which there are no data from lavas). The agreement between the secular-variation curve from the lake sediments, which has the advantage of being continuous, and the curve from the lavas, which should provide a set of accurate 'spot readings' of the geomagnetic field, suggests that there was a strong non-dipole field in Mexico about 5000 years BP that caused the declination to swing sharply to the east.

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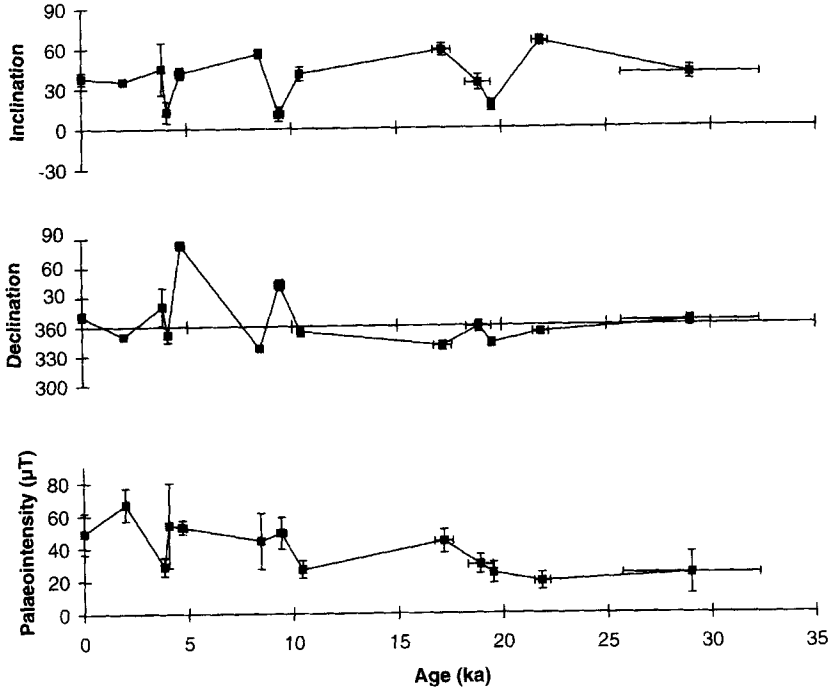


Figure 11. Secular variation curves for Central Mexico for the last 30 000 years, as defined by lavas from the Transmexican Volcanic Belt.

Over the 30 000 year period a gradual increase in palaeointensity may be observed, with shorter-period variations superimposed. On first inspection it might seem reasonable to attribute the longer-period variation to the main dipole field, and the shorter period variations to secular variation. However, to be certain of this, it is necessary to compare our palaeointensity data with data from the rest of the world. In order to do this, we need to remove the effect of latitude from the data, so the palaeointensities are combined with the inclinations to calculate virtual dipole moments (VDMs) (Smith 1967) (Table 7, Fig. 12).

VARIATIONS IN FIELD STRENGTH OVER THE LAST 30 000 YEARS

As variations in VDMs reflect changes not only in the dipolar but also the non-dipolar parts of the geomagnetic field, we would not expect complete agreement between the VDM records obtained from different parts of the globe. If there are contemporaneous highs or lows in the record, these may be interpreted as due to variation in the dipole field. However, if similar variations seem to be offset in time from place to place, this could be the result of the movement of a non-dipole source, i.e. a similar process to that which causes westward drift in directional records.

As the amount of palaeointensity data has grown it has become possible to look at variations in VDMs with time for the late Quaternary (in particular, the last 50 000 years). Since the study of Senanayake & McElhinny (1982), there have been a number of studies which have obtained further palaeointensity results for the last 50 000 years from ^{14}C-dated lavas (Tanaka 1990; Tanaka & Kono 1991; Mankinen & Champion 1993a,b; Tanaka *et al.* 1994). Compilations of these data indicate that there have been considerable variations in the strength of the dipolar part of the field. Tanaka (1991) has constructed curves of mean VDM data for Europe and Japan. For the past 9000 years, there are strong similarities between the 1000 year means from the two areas, with VDM maxima around 2000 and 8500 years BP. The addition of further data from Hawaii confirms the dipole high at around 9000 years BP (Mankinen & Champion 1993a,b). More recently, Tanaka *et al.* (1994) have obtained a very high palaeointensity from a 10 ka rhyolite flow in New Zealand, and this is interpreted by Mankinen (1994) as being a result of the dipolar peak being enhanced by a non-dipole field. The Hawaiian VDM record (Mankinen & Champion 1993a,b) also shows peaks at about 5000 and 14 000 years BP, which are interpreted as localized non-dipole variations.

Only materials containing a thermoremanent magnetization can yield absolute estimates of palaeointensity. However, lavas provide spot readings of the geomagnetic field, which are often difficult to date. Sediment cores, particularly from deep ocean sediments that formed under fairly constant conditions, can provide continuous records of the palaeomagnetic field, albeit somewhat smoothed. If the NRM intensities are normalized using ARM intensities or susceptibilities, it should be possible, as long as the magnetic mineralogy remains uniform throughout the core, to obtain estimates of the intensity of the geomagnetic field at the time that the NRMs formed. A number of authors have tried this in recent years (Constable & Tauxe 1987; Tauxe & Valet 1989; Tric *et al.* 1992; Meynadier *et al.* 1992; Valet & Meynadier 1993). The relative intensity curves show many features that are also observed in the VDM curves obtained from lavas. It is possible to use the palaeointensity record from lavas to calibrate these relative intensity curves and produce a pseudo-absolute record of the variation of the dipole moment [Meynadier *et al.* 1992 (Mediterranean and western Indian Ocean); Valet & Meynadier 1993 (Pacific Ocean)]. Comparison of their synthetic records, which show the variation of virtual axial dipole moments (VADM), with

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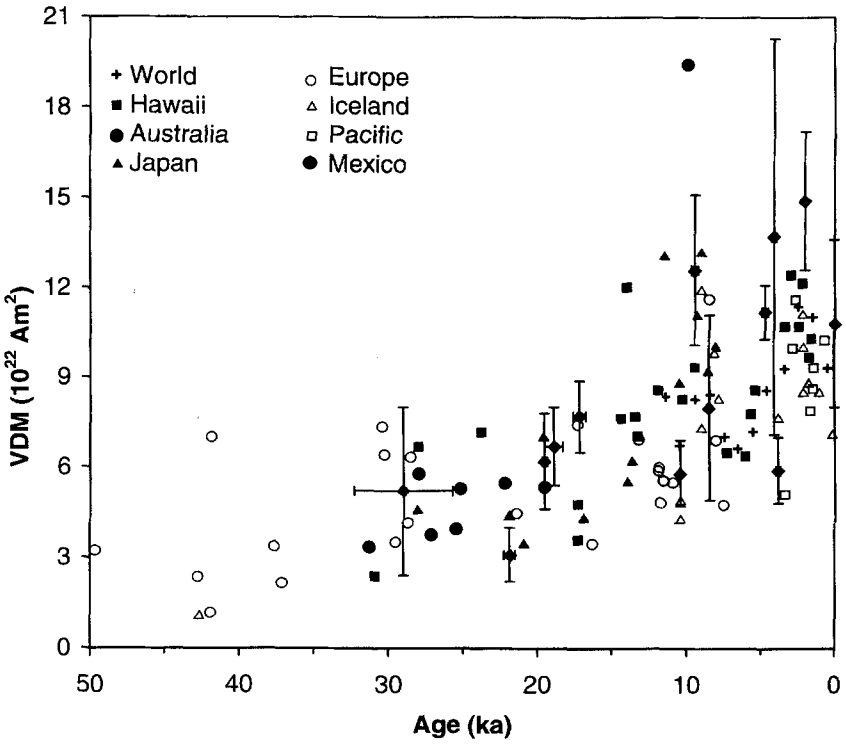


Figure 12. Graph showing VDMs for the last 50 000 years calculated from palaeointensity data from different parts of the world, together with the data obtained during this study. The solid line shows McElhinny & Senanayake's (1982) estimate of global VDMs for the last 10 000 years (after Schnepp 1991).

the VDM records is complicated by the fact that VADMs and VDMs are not directly comparable. As these authors had the relevant data available to them it is unclear why they did not calculate a synthetic virtual dipole moment, which uses the magnetic inclination, rather than a synthetic virtual axial dipole moment, which ignores variations in the magnetic inclination, and is therefore normally used for non-oriented archaeomagnetic samples. According to Tric *et al.* (1994), the differences between VADMs and VDMs are unlikely to exceed ± 20 per cent, unless the inclination is a very long way from the expected dipolar value. Comparison of the sedimentary synthetic VADM curve of Meynadier *et al.* (1992) and the VDM curve from lavas shows a good match for the last 45 000 years BP, and, in particular, the peak at 9000 years BP is clearly visible in both records (Mankinen 1994). From a low point at around 45 000 years BP, associated with the Laschamp excursion (Chauvin *et al.* 1989), a gradual increase in VDM, with a shorter-period signal superimposed, may be observed (Fig. 12).

Fig. 12 shows how the VDMs calculated from our Mexican lavas compare with VDMs obtained from late Quaternary rocks in other parts of the world. The new results fall within the same band as the earlier results. The high VDM ($14.9 \pm 2.3 \times 10^{22}$ Am^2) at about 2000 years BP from Xitle (S-9), is similar to the value reported for 1810 years BP by Tanaka & Kono (1991) from a Hawaiian lava. Mankinen & Champion (1993a) have produced further results from Hawaii for 2100 years BP which indicate a slightly lower value (12.15×10^{22} Am^2), and, in addition, throw doubt on the age of the flow measured by Tanaka & Kono (1991). The peak at about 5000 years BP observed in the Hawaiian record (Mankinen & Champion 1993a) also seems to be present in our data to some extent. A strong non-dipole source is the likely cause, as the declination at 4700 years BP (site M-6) was 83°. The high value at 9400 years BP (M-2) fits well with the peak in dipole field strength observed in both lavas and sedimentary records from the rest of the world. Our data from 17 000 to 19 500 years BP (sites M-10, S-3 and S-7) are in good agreement with one another, indicating a VDM of between 6 and 8×10^{22} Am^2 at that time. The data from the rest of the world are scattered (Mankinen 1994) for that period, suggesting that the variations in the non-dipole field were dominant at that time.

In order to obtain a better-constrained secular variation curve for Central Mexico, we need more points. There are many undated late Quaternary volcanoes in the Transmexican Volcanic Belt, and this study has shown that in many cases it is possible to obtain palaeomagnetic directions and palaeointensity estimates, so the major problem remains that of obtaining accurate dates for the lavas.

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